MILITARY SPECIFICATION

ELECTRODES, WELDING, COVERED, MOLYBDENUM
ALLOY STEEL APPLICATION

This amendment forms a part of Military Specification MIL-E-22200/7B, dated 28 March 1975, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Table I: Add "2/" to "Type 1/", and add as footnote 2/ the following:

"2/ The suffix RC added to the MIL-type pertains to restrictions of the copper content (see 3.4). All requirements, tests, and acceptance criteria remain otherwise unchanged."

2.1, under "STANDARDS": Add:

"FEDERAL
FED-STD-151 - Metals; Test Methods."

PAGE 2

3.3.1, line 1: After "deposited" add "and stress relieved".

PAGE 3

Table III, footnote 1/: Delete "note 6 or note 8" and substitute "note 7 or note 9".

Table IV: After the title add "2/", and add as footnote 2/ the following:

"2/ Use of the suffix RC (e.g., MIL-7010-A1 RC) indicates that the maximum copper content of the deposited weld metal shall be 0.15 percent. The remaining basic compositional requirements remain unchanged for specific MIL-type (see 6.2)."

Add as new paragraph 3.5.1:

"3.5.1 Heat and lot identification. When specified (see 6.2), each electrode shall be marked with heat or lot identification as specified in MIL-E-22200."

FSC 3439

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Add as new paragraph 3.7:

"3.7 Alloy identity. The alloy identity provisions of MIL-E-22200 shall be applied to each length of rod or wire to be spliced and to both ends of each wire coil drawn to finished size prior to cutting into electrodes. When specified (see 6.2), the core wire of each covered electrode shall meet the alloy identity requirements of MIL-E-22200."

PAGE 4

Table VII: Delete and substitute:

"TABLE VII. Quality conformance inspection.

Types	Tests	Procedures
All	No. 3 and No. 8a 1/ Table XI	Table IX Table XI
All with RC suffix	No. 3 and No. 8a 1/ plus a copper analysis Table XI	Table IX Table XI

1/ Impact properties shall be determined for quality conformance only when required by the acquisition document (see 6.2)."

PAGE 6

Table XI: Add the following:

"

Tests	Procedures	Requirements
Alloy identity	4.6.11 (MIL-E-22200)	3.6.4 (MIL-E-22200)

"

4.6: Delete and substitute:

"4.6 Certification of quality conformance. A certificate of quality conformance of the weld tests specified in 4.4.3 (i.e., X-ray, mechanical, and chemical analysis) containing actual quantitative test and inspection results, including the range of amperage employed in making the required welds for quality conformance inspection for each accepted lot included in a particular shipment, shall be furnished to the consignee with the shipment. The minimum quality conformance test result data required, including alloy identity results, shall be as shown on figure 3 and may be submitted in the format of figure 3."

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Add as new paragraphs 4.6.1 and 4.6.2:

"4.6.1 Identification of melts, heats, or lots. The identification of melts, heats, or lots, and the general requirements for test specimens shall be in accordance with FED-STD-151.

"4.6.2 Alloy identity certification. When specified (see 6.2), the inspector shall affix his stamp and date on the container containing electrodes conforming to MIL-E-22200. The letters "AT" shall be affixed to the container adjacent to the date to indicate "Alloy Tested"."

PAGE 7

Add as new paragraph 5.3.1.3:

"5.3.1.3 All containers shall include the following marking: "ASBESTOS FREE".

6.1: Delete and substitute:

"6.1 Intended use. Electrodes covered by this specification are intended for use in welded fabrication of piping, forgings, and castings made by molybdenum alloy steel and in the welding of composition C steel of MIL-S-23194 and MIL-S-24238. Degradation of high strength weld metal tensile and impact properties may occur because of extended exposure to stress relief temperatures and cooling from stress relief. These mechanical properties should be redetermined after simulated production heat treatment of sample material if required by applicable welding specifications and standards."

6.2: Add the following:

"(j) Whether the copper content is to be restricted as designated by the suffix RC to the MIL-type (see table IV).
"(k) Whether heat or lot identification is required for individual electrodes (see 3.5.1).
"(l) Whether the alloy identity test and certification is required for the core wire of each finished electrode (see 3.7 and 4.6.2)."

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PAGE 10

Figure 1, notes 7 and 9, item (a), line 2: Delete "1150° ± 25°F" and substitute "1125 ± 25°F".

PAGE 12

Figure 3: Delete and substitute the attached figure 3.

LAST PAGE

DD 1426, Standardization Document Improvement Proposal: Delete address and substitute:

"COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362"

Custodians:
Navy - SH
Air Force - 11

Review activities:
Air Force - 99
DLA - GS

User activities:
Army - AT
Navy - OS, YD

Preparing activity:
Navy - SH
(Project 3439-0401)

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CERTIFICATION OF QUALITY CONFORMANCE

Manufacturer or Distributor_____ Customer's Name_____
Address_____
Date_____ Customer's Order No._____
Specification MIL-_____ Core Wire Heat No._____
Type MIL-_____ Lot Identification MIL-E-22200 Para._____
Diameter and Length_____
Inspection Level_____ Wet Mix No._____
Lot No._____

Chemical Analysis (Complete)		Mechanical Test Stress-Relieved
	Yield strength (0.2 percent offset method)	
Carbon		
Manganese	Tensile strength	
Silicon	Elongation (percent)	
Sulfur	Reduction of area (percent)	
Phosphorus		
Molybdenum		
Copper (RC electrodes)		

Chemistry was taken from: ☐ Chem Pad ☐ Groove Weld

X-Ray: _____
Concentricity (percent)_____
Covering Moisture_____
Covering Iron Content_____

Groove Weld Test			
Test No.	3	8a	Chem Pad
Amperage			

Grinding During 8a Test Plate Preparation (see note 3) Operator Error (Layer Nos.) _____

Alloy identity tests: (a) Core wire_____
(b) Finished electrodes (if required, see 3.7)_____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____
Authorized company representative

NOTES:

1. Items not applicable to the type electrode involved or exempted by the acquisition document shall be marked N/A.
2. This form is applicable to the quality conformance tests required for each lot. A copy of the quality conformance test certification for the lot (figure 3) shall be forwarded with each shipment.
3. See 4.6 herein.

FIGURE 3. Certification of quality conformance.

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